

Parameter Robust Predictive Current Control for PMSM Drives Based on Self-Tuning Incremental Model and Voltage Constraint Compensation

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Abstract—Aiming to enhance the robustness and transient performance under model parameter mismatch, this article proposes a self-tuning incremental model-based predictive current control (STIM-PCC) strategy for permanent-magnet synchronous motor drives. In conventional incremental predictive current control scheme, although parameters such as permanent magnetic flux linkage and stator resistance are not required, stator inductance is still necessary for the predictive model. When the stator inductance mismatch occurs, prediction error is inevitable, leading to weakened robustness and deteriorated performance. To solve this issue, a novel STIM-PCC strategy is proposed, where the prediction error that indicates the inductance mismatch is adopted to tune the nominal inductance. In this way, the incremental model is updated adaptively and the precise predictive control can be achieved. Moreover, to enhance the tuning accuracy in the overmodulation region, a voltage constraint compensation method is put forward, which can effectively reduce the current drop during dynamic process. Compared to conventional scheme, the parameter robustness is significantly strengthened, where the inductance mismatch can be detected and corrected in time. Besides, not only the current fluctuation is reduced, but the settling time is shortened, thus greatly improving the transient performance. Experiments are carried out to validate the effectiveness of the proposed scheme.

Index Terms—Parameter robustness, permanent magnet synchronous motor (PMSM), predictive current control, self-tuning incremental model, transient performance.

I. INTRODUCTION

PERMANENT-MAGNET synchronous motors (PMSMs) are widely applied in various industrial fields like servo systems, electric vehicles, and more-electric aircraft because of their significant advantages of high-power density, high efficiency, and simple structure [1], [2], [3]. In these applications, rapid and accurate torque control is crucial to the high-quality operation of the system, where a fast current-loop is

required. However, it is difficult to obtain satisfactory performance through the traditional proportional-integral (PI) control scheme. Hence, various new current control methods have been developed, such as internal model control [4], sliding mode control [5], predictive control [6], [7], and deep-learning based control [8].

In recent years, deadbeat predictive current control (DPCC) has garnered considerable attention due to its excellent steady-state performance and fast transient response. However, due to the strong reliance on model parameters (e.g., permanent magnetic flux linkage, stator resistance and inductance), DPCC methods are sensitive and susceptible to the challenges posed by parameter mismatch. Consequently, once the parameters are mismatched, prediction error is unavoidable, leading to system performance deterioration and weakened robustness [9].

To address this challenge, several methods have been put forward to enhance the parameter robustness of DPCC scheme, which can be broadly categorized into three groups: parameters identification, extra compensation, and model improvement.

For the first type (i.e., parameters identification), there are two approaches to estimate the parameters: offline and online. Offline parameter identification involves determining the motor parameters before the control scheme is implemented, which can be done through experimental test [10] or techniques such as signal injection [11] and finite element analysis [12]. In comparison, online parameter estimation involves updating the motor parameters continuously during runtime, which is implemented by incorporating an algorithm or an observer [13], [14]. Online parameter estimation methods enable the control system to adapt to parameters variations over time, offering enhanced robustness against parameter mismatch. Therefore, online identification methods are generally preferred over offline methods. In [15], an identification model based on parameter perturbation method is established to estimate the parameter errors from the offsets of the DPCC. In [16], operating speed and inverter nonlinearity are considered, and parasitic transient tracking error of DPCC under inductance mismatch is adopted to identify the inductance. The offline and online identification methods are well reviewed in [17] and [18]. While these methods can enhance predictive model accuracy, challenges such as rank deficiency, computational resource costs, and model nonlinearity can constrain the identification performance. Consequently,

Received 20 May 2024; revised 10 October 2024 and 17 January 2025; accepted 24 January 2025. Date of publication 10 February 2025; date of current version 26 May 2025. This work was supported by the National Natural Science Foundation of China under Grant 52177043. Recommended for publication by Associate Editor S. C. Yang. (Corresponding author: Chun Gan.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPEL.2025.3539434>.

Digital Object Identifier 10.1109/TPEL.2025.3539434

the estimated results may not reliably converge to correct values. Besides, many of these algorithms are complex, and achieving accurate identification often necessitates a significant amount of training data.

In addition, to enhance the robustness of the system, it is common to introduce an extra compensation term into the predictive model. This compensation term is designed to counteract the negative effects caused by parameter mismatch and other interferences. In [19], a reduced-order generalized proportional-integral observer (GPIO) is utilized to track the disturbance caused by parameter mismatch. Through the GPIO compensation, anti-disturbance capability of the DPCC scheme is significantly improved. In [20], a sliding mode observer that integrates iterative learning control (ILC) is developed to observe disturbances arising from flux linkage distortion and dead time, thereby enhancing the prediction accuracy of the DPCC system. However, the incorporation of ILC introduces additional computational and storage burdens. In [21], parameter mismatches and load perturbation are incorporated as lumped disturbance and estimated using an extended state observer (ESO). In the control scheme, the structure “deadbeat + ESO” is applied in both speed and current control loop, making it hard to design the coefficients in the ESOs. Apart from the observers mentioned above, other types of observers are investigated and applied, such as Luenberger disturbance observer and extended high-gain state observer, which are comprehensively reviewed in [22] and [23]. However, the compensation terms are typically designed based on intricate principles, posing challenges for practical implementation. Besides, it is inevitable to introduce extra control parameters, where the tuning work would be complex and laborious.

The third approach to address the challenges arising from parameter mismatch is predictive model improvement. In [24] and [25], the sliding mode control theory, known for its strong robustness, is employed to rebuild the predictive model and mitigate the negative effects caused by parameter mismatch. In [26], active disturbance rejection control is merged with DPCC to improve control performance. Furthermore, the prediction model is trained using deep reinforcement learning, enhancing the model’s adaptation abilities. However, this approach leads to increased system complexity and computational burden, thereby constraining practical application. In [27] and [28], an ultralocal model is utilized to replace the machine model, and a model-free DPCC scheme is presented. Different from the approach in [27], which uses an ESO to estimate the unknown disturbance, in [28] the compensation term in the control loop and model gains are generated according to the data of voltage and current in the past control periods. The core concept behind the model-free methods is to bolster parameter robustness by removing the dependence on motor parameters in predictive model construction. While motor parameters are no longer necessary, the reliance on model parameters is actually shifted to the reliance on the high-precision disturbance observation. Additionally, extra parameters like observer gains are also introduced in the predictive models, which amplifies the complexity of the control system. Wang et al. [29] proposed the incremental model based predictive current control (IM-PCC) strategy. In this method, voltage

increment is predicted through the difference operation between adjacent control periods, thus eliminating the use of permanent magnetic flux linkage and resistance. Furthermore, in [30], [31], [32], the dynamic performance, prediction accuracy and stability margin of IM-PCC method are comprehensively investigated.

IM-PCC is based on the idea of iteration and no extra compensation or control parameters are introduced, which makes the system more robust and simpler. However, the inductance mismatch is ignored. To solve this issue, in [33] and [34], the concepts of multistep prediction and disturbance observer are introduced to provide a compensation term. However, more computation resources are required. In [35], an inductance updating method is provided, which is easy to implement. Nevertheless, it is derived by differentiating the DPCC model and thus cannot be directly applied to IM-PCC.

To address the issue of inductance reliance in the IM-PCC scheme, in this article, a self-tuning incremental model-based predictive current control (STIM-PCC) strategy is proposed. The main innovations of this article are the following points.

- 1) A diagnostic criterion for nominal inductance mismatch is established, where the prediction error that contains the information of inductance mismatch is revealed and adopted, so that the inductance mismatch can be quickly detected.
- 2) A novel self-tuning method is proposed. In this strategy, the actual current response that indicates the influence of the actual inductance is taken as the reference, hence the nominal inductance is tuned. On this basis, the incremental model are updated adaptively with the designed 2-step updating sequence, where precise predictive current control can be achieved.
- 3) Considering the special situation that voltage reference vector exceeds the modulation constraint, a voltage constraint compensation method is put forward, which can effectively reduce the current drop during dynamic process and enhance the tuning accuracy in such scenarios.

Besides, the cross-saturation phenomenon is illustrated and the impact on the proposed control strategy is detailly analyzed. Compared with the IM-PCC scheme, by using the proposed STIM-PCC scheme, no extra parameters are introduced, where the simple structure is maintained. Simultaneously, not only the parameter robustness can be significantly strengthened, but the transient performance is greatly improved.

II. CLASSIC PREDICTIVE CURRENT CONTROL FOR PMSM

In this section, first, the mathematical model of a PMSM is presented. Then, the conventional 2-step deadbeat predictive current control scheme is introduced, and based on it, the incremental predictive model is detailly derived and analyzed.

A. Mathematical Model of PMSM

In the d - q axis, the mathematical model of a PMSM is expressed by the following:

$$\begin{cases} u_d = R_s i_d + \frac{d\psi_d}{dt} - \omega_e \psi_q \\ u_q = R_s i_q + \frac{d\psi_q}{dt} + \omega_e \psi_d \end{cases} \quad (1)$$

$$\begin{cases} \psi_d = L_d i_d + \psi_f \\ \psi_q = L_q i_q \end{cases} \quad (2)$$

$$\begin{cases} T_e = \frac{3}{2} p i_q (\psi_f + (L_d - L_q) i_d i_q) \\ J \frac{d\omega_r}{dt} = T_e - T_l - B\omega_r \end{cases} \quad (3)$$

where u_d , u_q , i_d , i_q , ψ_d , and ψ_q are the d - q axis components of the stator voltage, current, and flux linkage. T_e , ω_r , and ω_e represent the electromagnetic torque, mechanical and electrical angular velocity, respectively. R_s , p , ψ_f , and J represent the stator resistance, number of rotor pole pairs, permanent-magnet flux linkage, and inertia constant, respectively. L_d and L_q are the d - q axis components of the stator inductance.

For PMSM drives, the two primary constraints that require attention are current and voltage limitation, which can be expressed in (4). In this equation, $I_{\max}$ and $U_{\max}$ represent the current and voltage constraints which are determined by the working conditions of converter and motor. Specially, $U_{\max}$ is generally set as $U_{dc}/1.732$ when the space vector pulsewidth modulation (SVPWM) method is employed

$$\begin{cases} i_d^2 + i_q^2 \leq I_{\max}^2 \\ u_d^2 + u_q^2 \leq U_{\max}^2 \end{cases} \quad (4)$$

B. 2-Step Deadbeat Predictive Current Control

Through the first-order forward Euler discretization, the voltage equation in (1) can be remodeled in a discrete form, where current at the next moment can be predicted according to the current and voltage at the present moment, as the following equation shows:

$$\begin{cases} i_d^{k+1} = \left(1 - \frac{T_s R_s}{L_d}\right) i_d^k + \frac{T_s}{L_d} u_d^k + \frac{T_s L_q \omega_e^k i_q^k}{L_d} \\ i_q^{k+1} = \left(1 - \frac{T_s R_s}{L_q}\right) i_q^k + \frac{T_s}{L_q} u_q^k - \frac{T_s \omega_e^k (L_d i_d^k + \psi_f)}{L_q} \end{cases} \quad (5)$$

where T_s is the sample period, and k is the sample index, representing the time instant $t = kT_s$. For the convenience of description, the signal sampled or applied at the k th moment is denoted with a superscript of k .

Deadbeat control involves determining the optimal control voltage that enables the current to track the current reference within single sampling period. Hence, by transforming (5), the optimal control voltage can be deduced as follows [33]:

$$\begin{cases} u_d^k = R_s i_d^k + L_d \frac{(i_d^k - i_{dref}^k)}{T_s} - \omega_e^k L_q i_q^k \\ u_q^k = R_s i_q^k + L_q \frac{(i_q^k - i_{qref}^k)}{T_s} + \omega_e^k (L_d i_d^k + \psi_f) \end{cases} \quad (6)$$

where u_{dref} and u_{qref} are the d - q axis components of the optimal control voltages and i_{dref} and i_{qref} are the current references.

However, in digital control application, there exist a time delay. Fig. 1 illustrates the updating sequence of PWM duty cycle. Due to the working principle of the digital signal processor (DSP), the voltage reference calculated at the k th control period cannot be immediately applied. Instead, the duty cycle is stored in the "shadow register" of the DSP module and applied in the subsequent $(k+1)$ th moment. As a result, the minimum control delay for tracking the current reference is two sampling periods, involving one period for current sampling and one period for implementing the voltage command.

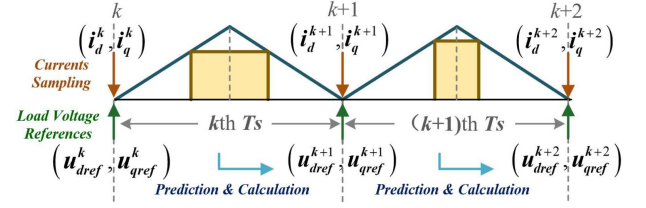


Fig. 1. Updating sequence of PWM duty cycle.

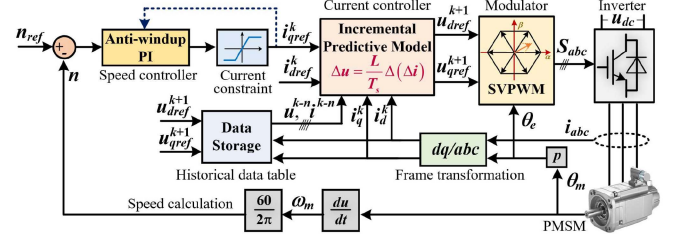


Fig. 2. Control structure of the IM-PCC scheme.

To address this problem, generally, delay compensation is utilized, where d/q -axis currents at the $(k+1)$ th moment are predicted by (5), hence the voltage references applied at the $(k+1)$ th moment are calculated by (6). However, this approach results in the doubling of computation costs. To alleviate the computational burden, the 2-step deadbeat predictive current control (2S-DPCC) scheme is proposed. In this scheme, the two sampling periods from the k th moment to the $(k+2)$ th moment are integrated as an entity, as demonstrated by the following:

$$\begin{cases} \frac{(u_d^k + u_{dref}^{k+1})}{2} = R_s \frac{(i_d^k + i_{dref}^k)}{2} + L_d \frac{(i_{dref}^k - i_d^k)}{2T_s} + E_d^k \\ \frac{(u_q^k + u_{qref}^{k+1})}{2} = R_s \frac{(i_q^k + i_{qref}^k)}{2} + L_q \frac{(i_{qref}^k - i_q^k)}{2T_s} + E_q^k \end{cases} \quad (7)$$

where E_d and E_q are the back electromotive force (EMF) items related to rotor speed

$$\begin{cases} E_d^k = -\omega_e^k L_q \frac{(i_q^k + i_{qref}^k)}{2} \\ E_q^k = \omega_e^k L_d \frac{(i_d^k + i_{dref}^k)}{2} + \omega_e \psi_f \end{cases} \quad (8)$$

Recollecting (7), voltage references can be obtained in the following:

$$\begin{cases} u_{dref}^{k+1} = R_s (i_d^k + i_{dref}^k) + L_d \frac{(i_{dref}^k - i_d^k)}{T_s} + 2E_d^k - u_d^k \\ u_{qref}^{k+1} = R_s (i_q^k + i_{qref}^k) + L_q \frac{(i_{qref}^k - i_q^k)}{T_s} + 2E_q^k - u_q^k \end{cases} \quad (9)$$

Through (9), delay compensation is not required. However, as is presented in (9), motor parameters such as R_s , ψ_f , L_d and L_q are utilized, where the prediction results can be deviated from the accurate value if the parameters are mismatched.

C. Incremental Model Based Predictive Current Control

To improve the parameter robustness and enhance the dynamic performance of 2S-DPCC, the incremental model based predictive current control (IM-PCC) strategy is put forward in [29], where the control structure is shown in Fig. 2. The incremental predictive model is derived as is following.

Similar to (7), the discretization equation during the interval from $(k-2)$ th moment to k th moment can be written as follows:

$$\begin{cases} \frac{(u_d^{k-2} + u_d^{k-1})}{2} = R_s \frac{(i_d^{k-2} + i_d^{k-1})}{2} + L_d \frac{(i_d^{k-2} - i_d^{k-1})}{2T_s} + E_d^{k-2} \\ \frac{(u_q^{k-2} + u_q^{k-1})}{2} = R_s \frac{(i_q^{k-2} + i_q^{k-1})}{2} + L_q \frac{(i_q^{k-2} - i_q^{k-1})}{2T_s} + E_q^{k-2} \end{cases} \quad (10)$$

Since the sampling frequency is much larger compared to the electrical frequency, the electrical velocity is approximated as constant during adjacent sampling periods. The back EMF items are roughly proportional to rotor speed, thus to simplify the design process, it can be assumed that $E^k \approx E^{k-1} \approx E^{k-2}$ [36].

On this basis, subtracting (10) from (7), the back EMF items can be eliminated, and the optimal voltage increment can be obtained, as presented in the following:

$$\begin{cases} \Delta u_{dref}^{k+1} = (u_d^k + u_{dref}^{k+1}) - (u_d^{k-2} + u_d^{k-1}) \\ = \left(R_s + \frac{L_d}{T_s}\right) (i_{dref}^k - i_d^{k-2}) - \frac{2L_d}{T_s} (i_d^k - i_d^{k-2}) \\ \Delta u_{qref}^{k+1} = (u_q^k + u_{qref}^{k+1}) - (u_q^{k-2} + u_q^{k-1}) \\ = \left(R_s + \frac{L_q}{T_s}\right) (i_{qref}^k - i_q^{k-2}) - \frac{2L_q}{T_s} (i_q^k - i_q^{k-2}) \end{cases} \quad (11)$$

Since R_s is much less than L/T_s , it can be neglected in (11). So (11) can be further simplified as follows:

$$\begin{cases} \Delta u_{dref}^{k+1} = \frac{L_d}{T_s} \Delta (i_{dref}^k) = \frac{L_d}{T_s} [(i_{dref}^k - i_d^k) - (i_d^k - i_d^{k-2})] \\ \Delta u_{qref}^{k+1} = \frac{L_q}{T_s} \Delta (i_{qref}^k) = \frac{L_q}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] \end{cases} \quad (12)$$

Recollect (12), the incremental predictive model is in (13).

$$\begin{cases} u_{dref}^{k+1} = \frac{L_d}{T_s} [(i_{dref}^k - i_d^k) - (i_d^k - i_d^{k-2})] - u_d^k + u_d^{k-1} + u_d^{k-2} \\ u_{qref}^{k+1} = \frac{L_q}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] - u_q^k + u_q^{k-1} + u_q^{k-2} \end{cases} \quad (13)$$

As is indicated in (13), in the IM-PCC method, through the difference operation between adjacent control periods, the use of R_s and ψ_f are eliminated, thus improving the robustness. Moreover, the incremental predictive model is much easier to be implemented than the 2S-DPCC model.

However, stator inductance is still necessary for the predictive model. According to the parameter sensitivity analysis presented in [32], the system will be unstable if the nominal inductance exceeds 125% of its actual value. In other words, the control system can only tolerate a mismatch in inductance of approximately 25%. When the stator inductance mismatch occurs, prediction error will be inevitable, leading to weakened robustness and deteriorated performance.

III. PROPOSED SELF-TUNING INCREMENTAL MODEL BASED PREDICTIVE CURRENT CONTROL STRATEGY

A novel self-tuning incremental model based predictive current control (STIM-PCC) strategy is proposed in this section, which can effectively enhance the parameter robustness and transient performance of the predictive current control. In this strategy, first, a diagnostic criterion is developed, where the inductance mismatch can be diagnosed by using the prediction error. Then, a self-tuning algorithm is proposed, through which the nominal inductance is tuned according to the actual current response and corresponding predictive current calculated by

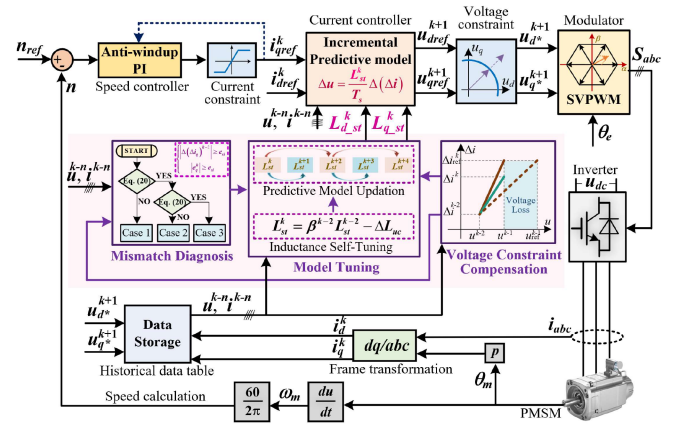


Fig. 3. Block diagram of the proposed STIM-PCC scheme.

historical currents. Based on the tuned inductance, the incremental model is updated adaptively with the designed updating sequence. Furthermore, in order to improve the tuning accuracy in the over-modulation region, a voltage constraint compensation method is put forward. The block diagram of the proposed STIM-PCC scheme is presented in Fig. 3.

In the diagram, the real-time motor speed is calculated based on the rotor position. PI controller is employed to design the speed controllers, and the back-calculation anti-windup method is utilized to mitigate the windup phenomenon, where the anti-windup gain is set as $1/K_p$. Switching signals are generated by using the SVPWM technique. As Fig. 3 shows, the proposed STIM-PCC scheme includes three submodules: mismatch diagnosis, model tuning, and voltage constraint compensation.

A. Inductance Mismatch Diagnosis

The accurate diagnosis of the inductance mismatch is fundamental to the tuning of predictive models. Therefore, it is necessary to develop a detection criterion that can identify whether the inductance is mismatched. Obviously, when the inductance mismatch occurs, the predictive error will be inevitable, which can be adopted as an indicator of inductance mismatch. Here, just taking q -axis for an example, according to (13), the q -axis current at the $(k+2)$ th moment can be obtained

$$i_q^{k+2} = \frac{T_s}{L_q} [(u_q^k + u_{qref}^{k+1}) - (u_q^{k-2} + u_q^{k-1})] + 2i_q^k - i_q^{k-2}. \quad (14)$$

Denoting the nominal q -axis inductance that is used in the predictive model as L_{qm} , then according to (12)

$$(u_q^k + u_{qref}^{k+1}) - (u_q^{k-2} + u_q^{k-1}) = \frac{L_{qm}}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})]. \quad (15)$$

Assuming that there is no error or deviation between the applied output voltage and the optimal voltage reference, that is, $u_q = u_{qref}$, then substitute (15) into (14)

$$\frac{L_q}{L_{qm}} (i_q^{k+2} - 2i_q^k + i_q^{k-2}) = i_{qref}^k - 2i_q^k + i_q^{k-2}. \quad (16)$$

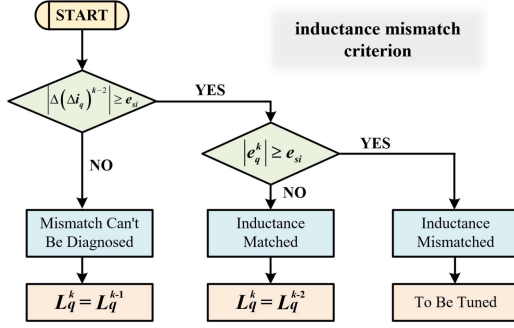


Fig. 4. Criterion for inductance mismatch diagnosis.

Define the prediction error e_q^{k+2} at the $(k+2)$ th moment as follows:

$$e_q^{k+2} = i_q^{k+2} - i_{qref}^k. \quad (17)$$

Let $L_{qm} = L_q + \Delta L_q$, then (18) can be deduced through (16)

$$e_q^{k+2} = \frac{\Delta L_q^k}{L_{qm}^k} (i_q^{k+2} - 2i_q^k + i_q^{k-2}) = \frac{\Delta L_q^k}{L_{qm}^k} \Delta(\Delta i_q)^k. \quad (18)$$

As is shown in (18), the relationship between the prediction error and the inductance mismatch can be revealed. First, it should be noted that, at the steady-state (i.e., $i_q^{k+2} = i_q^k = i_q^{k-2}$), the predictive current will not be affected by the inductance mismatch. Therefore, it can be inferred that the influence of inductance mismatch on the incremental predictive model primarily manifests in the dynamic process. On the other side, when the second-order current increment ($\Delta(\Delta i_q)$) is not zero, if the inductance is mismatched (i.e., $\Delta L_q \neq 0$), the prediction error e_q will be not zero. Thus, shifting (18) forward by 2 sampling periods, at the k th moment, the inductance mismatch of the $(k-2)$ th moment can be diagnosed as the following equation shows:

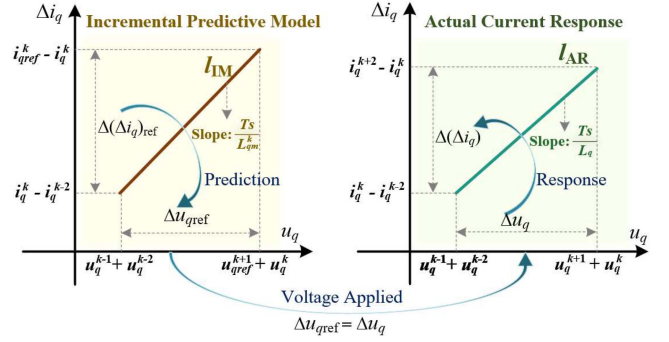
$$\begin{cases} \Delta(\Delta i_q)^{k-2} \neq 0 \\ e_q^k \neq 0 \end{cases} \Rightarrow \Delta L_q^{k-2} \neq 0. \quad (19)$$

In practical applications, the presence of various noise sources and interferences can introduce errors into the sampled current. Therefore, considering the impact of the sampling error, the inductance mismatch criterion can be rewritten as follows:

$$\begin{cases} |\Delta(\Delta i_q)^{k-2}| \geq e_{si} \\ |e_q^k| \geq e_{si} \end{cases} \Rightarrow \Delta L_q^{k-2} \neq 0 \quad (20)$$

where e_{si} represents the sampling error margin, accommodating current discrepancies resulting from various sources such as noise and external interferences. In practical applications, the range of e_{si} is determined by factors such as sensor error, noises and external disturbances, analog/digital conversion resolution, and voltage distortion. According to (20), the diagnosis process can be visualized as presented in Fig. 4.

As indicated in Fig. 4, the nominal inductance is determined by the most accurate value at that instant. Specifically, when the inductance is detected to be matched, signifying the accuracy of L^{k-2} , thus the nominal inductance for the current control period will be set as L^{k-2} (i.e., $L^k = L^{k-2}$). Conversely, if the inductance

Fig. 5. Relationship between the incremental predictive model and the actual current response (from the viewpoint of u_q - Δi_q plane).

mismatch is detected, the accurate value will be obtained through the self-tuning algorithm, allowing for the adjustment of the nominal inductance and the L^k can be tuned and set. If the parameter mismatch goes undiagnosed, the accuracy of L^{k-2} is hard to distinguish. In such cases, the nominal inductance used in the previous control period (L^{k-1}) is regarded as the most precise value at that moment, and is retained for the current control period (i.e., $L^k = L^{k-1}$).

B. Self-Tuning Algorithm for Incremental Predictive Model

Thanks to the diagnosis criterion introduced in part A, the inductance mismatch can be detected in time. Then, in this part, a self-tuning algorithm is proposed to tune the nominal inductance, and based on it, the incremental predictive model is updated through the designed updating sequence. For the convenience of description, q -axis is still taken for an example. The details are analyzed in the following.

1) *Tuning of the Mismatched Nominal Inductance:* As is indicated in (14) and (15), the optimal voltage increment is generated through the predictive model according to the current reference, and subsequently, applied in the actual system, where the actual current response is triggered. Therefore, the optimal voltage increment serves to establish a connection between the predictive model and the actual response, which represent the impact of the nominal inductance and the actual inductance, respectively, as is illustrated in the following:

$$\begin{cases} \underbrace{(u_q^k + u_{qref}^{k+1}) - (u_q^{k-2} + u_q^{k-1}) = \frac{L_{qm}^k}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})]}_{\text{Incremental predictive model}} \\ \underbrace{(i_q^{k+2} - i_q^k) - (i_q^k - i_q^{k-2}) = \frac{T_s}{L_q} [(u_q^k + u_q^{k+1}) - (u_q^{k-2} + u_q^{k-1})]}_{\text{Actual current response}} \end{cases} \quad (21)$$

As (21) presents, the equations depict a linear relationship between the 2nd-order current increment and the voltage increment. Consequently, the equations in (21) can be represented as lines in the u_q - Δi_q plane. In this way, assuming that there is no error or deviation between the u_q and u_{qref} (i.e., $u_q = u_{qref}$), the connection between the incremental predictive model and the actual current response can be shown in Fig. 5.

In Fig. 5, the corresponding lines of the incremental predictive model and the actual current lines in the u_q - Δi_q plane are denoted as l_{IM} and l_{AR} . It is evident that when there is a mismatch in the inductance, the slopes of l_{IM} and l_{AR} will be different. Therefore, the line l_{AR} , characterized by a slope of T_s/L_q (which contains the information of the actual inductance), can serve as the tuning reference. Then, the line l_{IM} , initially having a slope of T_s/L_{qm}^k , can be adjusted to align with the line l_{AR} through modifying its slope. In this way, the mismatched nominal inductance can be tuned.

Substituting $u_q = u_{qref}$ into (21), then (22) can be deduced:

$$\frac{L_q}{T_s} [(i_q^{k+2} - i_q^k) - (i_q^k - i_q^{k-2})] = \frac{L_{qm}^k}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})]. \quad (22)$$

Recollecting (22), then the actual inductance can be calculated by the nominal inductance, as the following shows:

$$\frac{L_q}{L_{qm}^k} = \frac{(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})}{(i_q^{k+2} - i_q^k) - (i_q^k - i_q^{k-2})} = \frac{\Delta(\Delta i_q)^k_{ref}}{\Delta(\Delta i_q)^k}. \quad (23)$$

Based on (23), the q -axis inductance can be tuned under the knowledge of the i_q^{k+2} . However, at the k th moment, the most recent available current is i_q^k . Therefore, to address this limitation, (23) is shifted forward by 2 sampling periods. In this way, the nominal inductance used at the $(k-2)$ th moment and the currents from the $(k-4)$ th moment to the k th moment are employed to tune the nominal inductance. Consequently, the tuning algorithm is formulated as (24).

$$L_{q-st}^k = \beta^{k-2} L_{q-st}^{k-2} = \frac{\Delta(\Delta i_q)^{k-2}_{ref}}{\Delta(\Delta i_q)^{k-2}} L_{q-st}^{k-2} = \frac{i_{qref}^{k-2} - 2i_q^{k-2} + i_q^{k-4}}{i_q^k - 2i_q^{k-2} + i_q^{k-4}} L_{q-st}^{k-2} \quad (24)$$

where β is the self-tuning coefficient, and to distinguish with the nominal inductance, the tuned inductance is marked as L_{q-st} .

To provide a more intuitive demonstration for the effect of inductance tuning, two cases of nominal inductance mismatch are considered (initial nominal inductance is greater/smaller than the actual inductance). Additionally, the subsequent predictions are presented, as depicted in Fig. 6.

In Fig. 6, $\Delta u_{q,t}$ and $\Delta u_{q,n}$ represent the predictive voltage increments at the $(k+1)$ th moment, which are calculated with the inductance tuned and not tuned, respectively. When the initial nominal inductance is smaller than the actual inductance (i.e., case in Fig. 6(a)), the slope of the line l_{IM} is greater than that of the line l_{AR} , and according to (15), this results in the optimal control voltage being smaller than that calculated with matched inductance. In this case, the inductance will be tuned to a higher value, where the predictive voltage increment $\Delta u_{q,t}$ will be greater than the untuned situation so that the current at the $(k+2)$ th moment can track the current reference. Similarly, when the initial nominal inductance is greater than the actual inductance (i.e., case in Fig. 6(b)), the slope of the line l_{IM} is smaller than that of the line l_{AR} . In this case, the inductance will be tuned to a smaller value, where the $\Delta u_{q,t}$ will be smaller than the untuned situation so that the drastic current responses resulting from excessive voltage variation can be prevented.

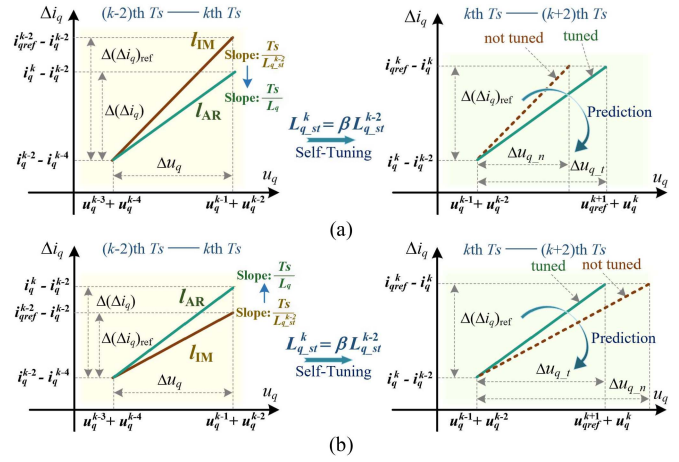


Fig. 6. Prediction performance comparison with the inductance tuned and not tuned. (a) Initial nominal inductance is smaller than the actual inductance. (b) Initial nominal inductance is greater than the actual inductance.

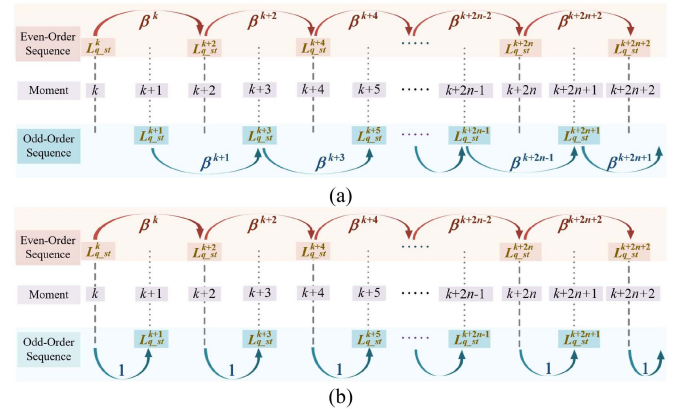


Fig. 7. Updating sequences of the tuned inductance. (a) Updating sequence that even-order sequence and the odd-order sequence are independent of each other. (b) Proposed 2-step updating sequence.

2) *Updating of the Predictive Model:* In accordance with the tuning algorithm outlined in (24), the inductance at the k th moment is tuned based on the value adopted two sampling periods ahead. Therefore, assuming that inductance tuning commences at the k th moment, the sequence of inductance tuning can be deduced as (25) and (26).

For the following odd-numbered moments

$$L_{q-st}^{k+2n+1} = \beta^{k+2n-1} L_{q-st}^{k+2n-1} = \dots = \beta^{k+2n-1} \dots \beta^{k+1} L_{q-st}^{k+1}. \quad (25)$$

For the following even-numbered moments

$$L_{q-st}^{k+2n} = \beta^{k+2n-2} L_{q-st}^{k+2n-2} = \dots = \beta^{k+2n-2} \dots \beta^k L_{q-st}^k. \quad (26)$$

where n is a positive integer used for moment counting.

It can be observed from (25) and (26) that the even-order sequence and the odd-order sequence are independent of each other. In this case, if the nominal inductance is tuned at every moment, as is shown in Fig. 7(a), both sequences will converge to the same value (L_q). This results in unnecessary repetition of the tuning process. To address this issue and simplify the updating sequence, a 2-step updating sequence is proposed, as Fig. 7(b)

shows. In this method, the even-order sequence is maintained, while the tuned inductance is held as the nominal inductance for the next odd-order moment.

Based on the tuned inductance, the incremental predictive model can be updated adaptively, as is illustrated in the following:

$$\begin{cases} u_{dref}^{k+1} = \frac{L_{d, st}^k}{T_s} [(i_{dref}^k - i_d^k) - (i_d^k - i_d^{k-2})] - u_d^k + u_d^{k-1} + u_d^{k-2} \\ u_{qref}^{k+1} = \frac{L_{q, st}^k}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] - u_q^k + u_q^{k-1} + u_q^{k-2}. \end{cases} \quad (27)$$

Consider the voltage limitation as (4) shows, the magnitude of voltage reference will be limited within the constraint:

$$u_{d, q*}^{k+1} = \begin{cases} u_{d, qref}^{k+1} \left(\sqrt{(u_{dref}^{k+1})^2 + (u_{qref}^{k+1})^2} \leq U_{max} \right) \\ \frac{U_{max} u_{d, qref}^{k+1}}{\sqrt{(u_{dref}^{k+1})^2 + (u_{qref}^{k+1})^2}} \left(\sqrt{(u_{dref}^{k+1})^2 + (u_{qref}^{k+1})^2} > U_{max} \right) \end{cases} \quad (28)$$

where $u_{d, q*}$ represent the constrained optimal voltage references that is used in the SVPWM module.

C. Voltage Constraint Compensation for O-M Region

The aforementioned analysis in Sections II-A and B is predicated on a fundamental assumption: the actual voltage can be effectively modulated to match the optimal voltage reference (i.e., $u_q = u_{qref}$). In other words, the optimal voltage reference vector is presumed to fall within the linear-modulated (L-M) region.

However, there are certain scenarios, such as a large step signal is applied to the current reference, where the optimal voltage reference vector may lie outside the L-M region and instead fall within the overmodulation (O-M) region. In such cases, the optimal voltage references are subject to constraints, as (28) shows, rendering the assumption of $u_q = u_{qref}$ invalid. Therefore, for O-M region, it is necessary to optimize the inductance mismatch diagnosis method and tuning algorithm.

First, when $u_q \neq u_{qref}$, the accuracy of the inductance mismatch diagnosis will be compromised. To solve this issue, denote the voltage loss resulting from the voltage constraint as follows:

$$u_{qloss} = u_{qref} - u_q. \quad (29)$$

According to (14), (15), and (29), (16) can be rewritten as follows:

$$\frac{L_q}{L_{qm}^k} (i_q^{k+2} - 2i_q^k + i_q^{k-2}) = i_{qref}^k - 2i_q^k + i_q^{k-2} - \frac{T_s u_{qloss}^{k+1}}{L_{qm}^k} \quad (30)$$

Based on (30), it can be deduced that

$$i_q^{k+2} - i_{qref}^k + \frac{T_s u_{qloss}^{k+1}}{L_{qm}^k} = \frac{\Delta L_q}{L_{qm}^k} (i_q^{k+2} - 2i_q^k + i_q^{k-2}). \quad (31)$$

As is presented in (31), considering the voltage constraint, the prediction error e_q^{k+2} is redefined as follows:

$$e_q^{k+2} = i_q^{k+2} - i_{qref}^k + \frac{T_s u_{qloss}^{k+1}}{L_{qm}^k}. \quad (32)$$

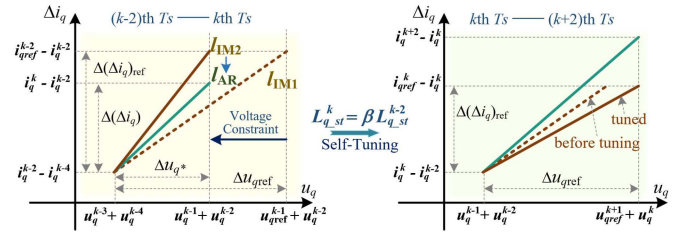


Fig. 8. Inductance tuning with the assumption of $u_q = u_{qref}$ in the O-M region.

By redefining e_q^{k+2} , (18) can still hold, ensuring the validity of the inductance mismatch criterion shown in (20). As a result, the diagnosis accuracy in the O-M region can be guaranteed.

Additionally, when the voltage references are constrained, the tuning accuracy of the mismatched inductance will be diminished. Fig. 8 depicts the tuning process under the assumption of $u_q = u_{qref}$ in the O-M region. In this figure, the respective lines of the incremental predictive model and the actual current response in u_q - Δi_q plane are l_{IM1} and l_{AR} . Initially, the nominal inductance is set to be larger than the actual value, resulting in a smaller slope for l_{IM1} compared to l_{AR} .

As shown in Fig. 8, the optimal voltage reference u_{qref}^{k-1} is generated based on the nominal inductance at the $(k-2)$ th moment and is constrained to u_q^{k-1} . It can be inferred from (24) that, if the tuning algorithm is directly employed with the assumption of $u_q = u_{qref}$, the line l_{IM2} , which shares the same “ Δu_q ” value with l_{AR} , would be mistakenly treated as the predictive model. In this scenario, l_{IM2} is adjusted to align with l_{AR} by reducing its slope, that is, increasing the nominal inductance. Consequently, this process leads to a greater deviation in inductance, ultimately triggering a more significant current response in subsequent control period, as depicted in Fig. 8.

To address this issue and improve the tuning accuracy in the O-M region, a voltage constraint compensation for inductance tuning is proposed. By substituting (29) into (21), then

$$\frac{L_q}{T_s} (i_q^{k+2} - 2i_q^k + i_q^{k-2}) = \frac{L_{qm}^k}{T_s} (i_{qref}^k - 2i_q^k + i_q^{k-2}) - u_{qloss}^{k+1}. \quad (33)$$

Recollecting (33), then (34) can be derived

$$\frac{L_q}{L_{qm}^k} = \frac{\Delta (\Delta i_q)_{ref}^k}{\Delta (\Delta i_q)^k} - \frac{T_s u_{qloss}^{k+1}}{\Delta (\Delta i_q)^k}. \quad (34)$$

Shifting (34) forward by two sampling periods, the tuning algorithm with the voltage compensation is formulated as follows:

$$\begin{aligned} L_{q, st}^k &= \beta^{k-2} L_{q, st}^{k-2} - \Delta L_{q, uc} \\ &= \frac{\Delta (\Delta i_q)_{ref}^{k-2}}{\Delta (\Delta i_q)^{k-2}} L_{q, st}^{k-2} - \frac{T_s u_{qloss}^{k+1}}{\Delta (\Delta i_q)^{k-2}} \\ &= \frac{i_{qref}^{k-2} - 2i_q^{k-2} + i_q^{k-4}}{i_q^k - 2i_q^{k-2} + i_q^{k-4}} L_{q, st}^{k-2} - \frac{T_s u_{qloss}^{k+1}}{i_q^k - 2i_q^{k-2} + i_q^{k-4}} \end{aligned} \quad (35)$$

where $\Delta L_{q, uc}$ is the compensation term. Similarly, the tuning of the nominal d -axis inductance can also be improved to compensate for the voltage constraint, which is omitted here.

D. Analysis on the Impact of Cross-Saturation

Specially, in the actual physical PMSM model, there exists a common magnetic circuit between d -axis and q -axis. This feature causes the cross-saturation effect and the mutual inductance between d -axis and q -axis [37], [38]. Usually, the cross-saturation inductance is much smaller than the self-inductance, thereby the cross-saturation is often ignored in most research works studying DPCC. Similarly, in the proposed method, only d/q -axis self-inductance is considered and tuned. To provide a comprehensive view, in this part, the analysis on the cross-saturation effect is provided.

Considering the cross-saturation phenomenon, the d/q -axis stator flux linkage can be rewritten as follows:

$$\begin{cases} \psi_d = L_{dd}i_d + L_{dq}i_q + \psi_f \\ \psi_q = L_{qq}i_q + L_{qd}i_d \end{cases} \quad (36)$$

where L_{dd} and L_{qq} are the d/q -axis self-inductance, that is, the “ L_d ” and “ L_q ” mentioned above. L_{dq} and L_{qd} are the cross-saturation inductances. On this basis, when the cross-saturation inductances are considered, the incremental form of the optimal voltage references can be deduced as follows:

$$\begin{cases} u_{dref}^{k+1} = \frac{L_{dd}}{T_s} [(i_{dref}^k - i_d^k) - (i_d^k - i_d^{k-2})] - u_d^k + u_d^{k-1} + u_d^{k-2} \\ + \frac{L_{dq}}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] \\ u_{qref}^{k+1} = \frac{L_{qq}}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] - u_q^k + u_q^{k-1} + u_q^{k-2} \\ + \frac{L_{qd}}{T_s} [(i_{dref}^k - i_d^k) - (i_d^k - i_d^{k-2})] \end{cases} \quad (37)$$

As indicated in (37), leveraging the voltages in neighboring control periods, in the steady-state ($i_{ref}^k \approx i^k \approx i^{k-2}$), the influence of cross-saturation inductances can be neutralized.

For current dynamic response, a q -axis current variation is assumed (i.e., the q -axis current relevant differential term in (37) are not zero). In this case, (37) can be further deduced as follows:

$$\begin{cases} u_{dref}^{k+1} = \frac{L_{dq}}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] - u_d^k + u_d^{k-1} + u_d^{k-2} \\ u_{qref}^{k+1} = \frac{L_{qq}}{T_s} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] - u_q^k + u_q^{k-1} + u_q^{k-2} \end{cases} \quad (38)$$

Comparing (13) and (38), when a q -axis current variation is applied, it is apparent that the functions of the optimal q -axis voltage reference in both models are the same. Thus, with the proposed self-tuning method, the detection and tuning for the mismatched L_{qq} is still effective. However, an extra term is introduced into the d -axis voltage reference. Specifically, at the $(k+2)$ th instant, the d -axis current error can be deduced as follows:

$$e_d^{k+2} = i_d^{k+2} - i_{dref}^k = -\frac{L_{dq}}{L_{dd}} [(i_{qref}^k - i_q^k) - (i_q^k - i_q^{k-2})] \quad (39)$$

As presented in (39), due to the cross-saturation inductance, a current tracking error will be caused in d -axis current when a q -axis current variation is applied. The cross-inductance is typically smaller than the self-inductance, thereby confining the d -axis current error within the error margin established in the mismatch detection submodule. In this way, the detection and tuning for L_{dd} will not be interfered. Notably, to completely

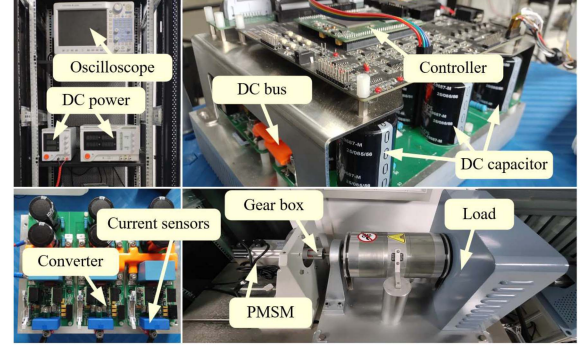


Fig. 9. Experimental platform.

TABLE I
PMSM PARAMETERS

Parameters	Unit	Value
Pole pairs	-	5
Permanent magnet flux linkage	Wb	0.0473
d -axis inductance	mH	3.45
q -axis inductance	mH	4.2
Stator resistance	Ω	0.9
Current constraint	A	4.8
Voltage constraint	V	170
Rated speed	rpm	1000
Rated load torque	Nm	1

mitigate the influence from the cross-inductance, it necessitates enlarging the error margin so that larger current variation can be tolerated, which may compromise the tuning accuracy.

Therefore, the steady-state performance with the proposed STIM-PCC scheme is not affected by the cross-saturation inductance. While in the dynamic process, a tracking error can be introduced, whose impact can be isolated through the error margin design to forbid error-detection and error-tuning.

IV. EXPERIMENTAL VERIFICATION

The effectiveness and performance of control methods are evaluated experimentally on a three-phase PMSM test platform as shown in Fig. 9, and the main motor parameters are shown in Table I. In this platform, a dc power supply with constant output is employed to provide power to the motor system. A converter is used for motor control. The control algorithm is implemented based on a DSP TMSF28377 control board. The phase currents are detected by current sensors LA-55P/SP1. A magnetic brake is adopted as the load.

The fundamental settings for the following experiments are as below: the sampling frequency and the switching frequency of the inverter are 10 kHz, and the dc bus voltage is 310 V. The load torque is 1 N·m. All the signals including the rotor speed, the d/q -axis currents, q -axis voltage are displayed via an on-board D/A converter. A multichannel isolated oscilloscope Yokogawa DL850E is used to record the waveforms.

In the experiments, the control loop is constructed based on the block diagram shown in Fig. 3. It should be stated that the same PI coefficients (applied in the speed controller) are utilized in all experiments below to ensure fair comparisons.

TABLE II
CASES FOR ANALYSIS OF INDUCTANCE MISMATCH

Case	1	2	3	4
Nominal inductance	3.0mH	5.0mH	5.5mH	4.2mH
Degree of mismatch	30% ↓	20% ↑	30% ↑	matched
Stability	stable	stable	unstable	stable

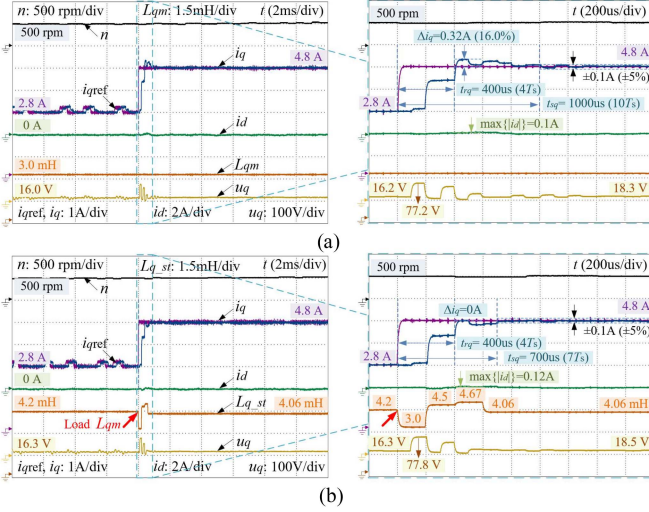


Fig. 10. q -axis current step response with inductance mismatch when $L_{qm} = 3$ mH ($\approx 0.7L_q$). (a) IM-PCC method. (b) Proposed STIM-PCC method.

A. Transient Performance Verification

In this part, two typical dynamic processes are examined to verify the transient performances of the control system. First, the current step response under inductance mismatch is utilized to illustrate the current transient performance and parameter robustness. Besides, the speed response during sudden load changes is tested to present the speed transient performance.

1) *Current Step Responses With Inductance Mismatch*: The condition of the current step response test is that the motor runs at 500 r/min with rated load (1 N·m), and the current reference steps from 2.8 A (corresponding current under the rated load) to 4.8 A (the maximum current). When adopting the conventional IM-PCC method, the nominal inductance is set as the specified value. When employing the proposed STIM-PCC method, in order to depict the tuning process of the nominal inductance, the nominal inductance will be set as the same specified value at the “step instant”, and then be adjusted through the tuning algorithm. For the convenience of comparison, response of q -axis current is taken for the example.

According to the parameter sensitivity analysis outlined in [32], the control system with the IM-PCC method will be unstable if the nominal inductance exceeds 125% of its actual value. To provide a comprehensive demonstration of the influence of inductance mismatch, four cases are considered, where the applied nominal q -axis inductance is intentionally mismatched to different degrees as is indicated in Table II.

Fig. 10 presents the experimental results for the first case. In the figure, the current overshoot (Δi_q), 90% rising time (t_r), and 5% settling time (t_s) are utilized as key indicators for transient performance analysis. As shown in Fig. 10, when the initial nominal inductance is smaller than the actual value, the

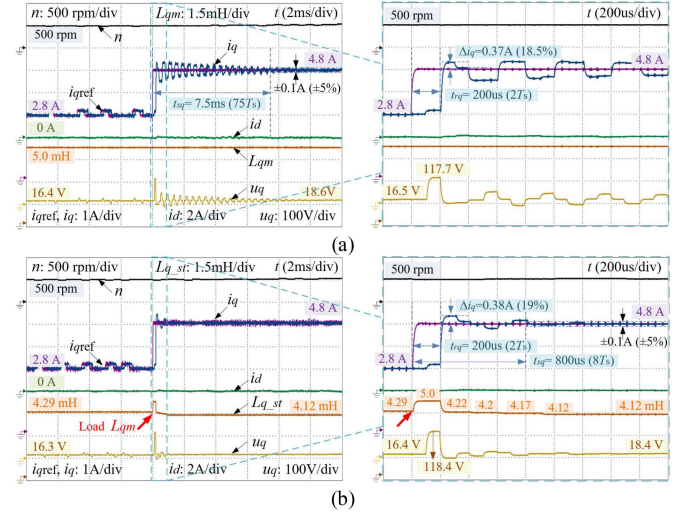


Fig. 11. q -axis current step response with inductance mismatch when $L_{qm} = 5$ mH ($\approx 1.2L_q$). (a) IM-PCC method. (b) Proposed STIM-PCC method.

proposed STIM-PCC strategy demonstrates superior dynamic performance compared to the conventional IM-PCC strategy. With the conventional IM-PCC strategy, noticeable overshoot is observed ($\Delta i_q = 0.32$ A, 16% of the current step), and it takes ten switching periods to settle the current ($t_s = 10T_s$). By using the proposed STIM-PCC strategy, the inductance mismatch is quickly detected and adjusted at the second moment after setting the mismatched inductance, thus the current overshoot can be effectively eliminated. The settling time is also reduced significantly compared to the conventional method, with only $7T_s$ required to settle the current. Besides, as the tuning process in Fig. 10(b) shows, the mismatched inductance can be tuned in just $6T_s$, with a relative error of only 3.3%, demonstrating the high accuracy of the proposed self-tuning algorithm.

Additionally, it can be observed that, in both cases, slight d -axis current fluctuation occurs during the q -axis current step process, and as soon as the q -axis current tracks the reference, the d -axis current fluctuation vanishes. This phenomenon is caused by the cross-saturation, as is analyzed in part D of Section III. Since the d -axis current variation is too slight (at maximum of 0.12 A, within the error margin), the detection and tuning for d/q -axis inductance is not disturbed.

Fig. 11 presents the experimental results for case 2, where the mismatched inductance is set as 5 mH (approximately 20% larger than the actual value).

As is presented in Fig. 11, when applying the conventional IM-PCC method, the q -axis current follows the reference with long-term fluctuations, and it takes 75 switching periods to settle the current ($t_s = 75T_s$). However, with the proposed STIM-PCC strategy, the current settling time is significantly reduced, requiring only $8T_s$ (a reduction of $67T_s$). Besides, precise tuning is achieved, with the mismatched inductance being tuned in just $9T_s$ and a relative error of only 1.9%.

In cases 1 and 2, the current control with the conventional IM-PCC strategy can remain stable despite the inductance mismatches, as is demonstrated in Figs. 10 and 11. However, in case 3, an inductance mismatch of 30% is introduced, which

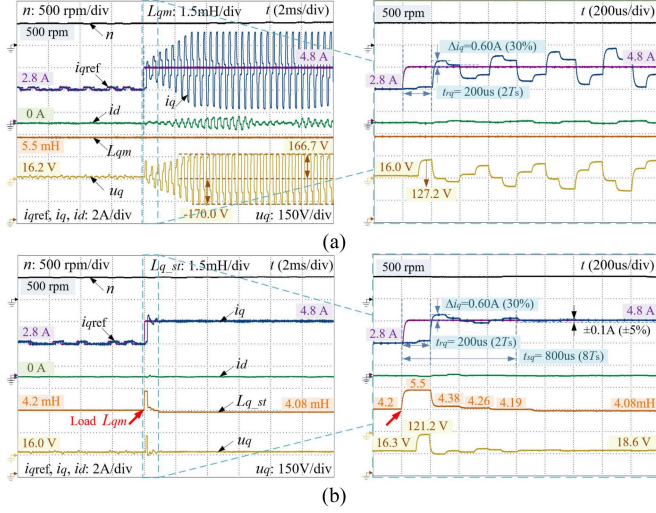


Fig. 12. q -axis current step response with inductance mismatch as $L_{qm} = 5.5$ mH ($\approx 1.3L_q$). (a) IM-PCC method. (b) Proposed STIM-PCC method.

exceeds the aforementioned stability margin (25%), bringing great challenge to the stable control of current.

Fig. 12 displays the experimental results for case 3.

It can be observed in Fig. 12 that, with the conventional IM-PCC strategy, sharp current fluctuations occur following the reference step. In this process, the q -axis voltage is saturated rapidly, resulting in continuous constant oscillations in the q -axis current, rendering the q -axis current uncontrollable. Simultaneously, great d -axis current fluctuation is triggered due to the cross-saturation effect. However, with the proposed STIM-PCC method, the inductance mismatch is quickly diagnosed and tuned within $9T_s$, where high tuning accuracy is also achieved with a relative error of only 2.9%. Moreover, stable current control is realized with the proposed control strategy, where only $8T_s$ are required to settle the current. At the same time, d -axis current variation caused by cross-saturation is slight, thanks to the improved q -axis current response. Thus, compared to the conventional method, the proposed strategy significantly enhances parameter robustness and transient performance of the predictive current control.

When the inductance is matched (i.e., case 4), experiments are also conducted using both methods, as shown in Fig. 13.

As is displayed in Fig. 13, when the nominal inductance is matched, the tuning algorithm would not come into effect, resulting in identical current responses with both methods. In this case, the best current transient performance can be achieved, where the current can rapidly track the reference value with minimal fluctuations, settling within just six periods.

According to the indicators including Δi_q , t_r , and t_s , the current transient performances of all cases with both methods can be compared, which are summarized in Table III.

Table III demonstrates the significant improvement in transient performance and parameter robustness achieved by employing the proposed STIM-PCC strategy. A comparison with the conventional IM-PCC method reveals that the proposed strategy leads to reduced current fluctuations during the dynamic process and significantly shortened settling time. Furthermore,

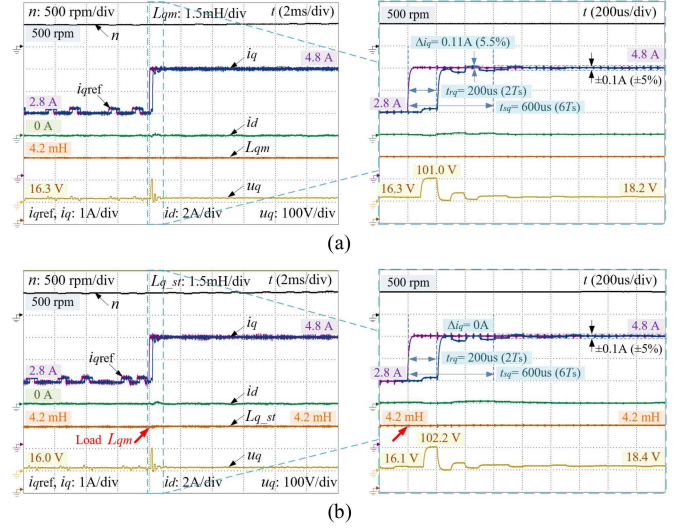


Fig. 13. q -axis current step response without inductance mismatch. (a) IM-PCC method. (b) Proposed STIM-PCC method.

TABLE III
CURRENT TRANSIENT PERFORMANCE COMPARISON

Case	Method	Δi_q (A)	t_r	t_s	ΔL_{qs}
$L_{qm} \approx 0.7L_q$	IM-PCC	0.32	$4T_s$	$10T_s$	-28.5%
	STIM-PCC	0	$4T_s$	$7T_s$	-3.3%
$L_{qm} \approx 1.2L_q$	IM-PCC	0.37	$2T_s$	$75T_s$	19%
	STIM-PCC	0.38	$2T_s$	$8T_s$	1.9%
$L_{qm} \approx 1.3L_q$	IM-PCC	0.6	$2T_s$	unstable	31%
	STIM-PCC	0.6	$2T_s$	$8T_s$	2.9%
$L_{qm} = L_q$	IM-PCC	0.11	$2T_s$	$6T_s$	0
	STIM-PCC	0	$2T_s$	$6T_s$	0

under the inductance mismatches of different degrees, the nominal inductance can be swiftly tuned to match the actual value within ten switching periods, with maintaining high accuracy (tuning errors are smaller than 5% in all cases).

Similarly, since the incremental models in d -axis and q -axis are symmetrical, the proposed STIM-PCC method is definitely effective for the situations with mismatched L_d . To prove it, experiments are also conducted, the detailed results and relevant explanations are shown in part A in the appendix.

Besides, the self-tuning performance across various current sampling frequencies are also investigated and experiments are carried out too. Details are presented in Appendix B.

2) *Significance of the Voltage Constraint Compensation:* To clearly illustrate the effectiveness and significance of the voltage constraint compensation method, experiments with the STIM-PCC method are conducted, where the optimal voltage reference vector falls within the O-M region, as Fig. 14 depicts. The condition of the tests shown in Fig. 14 is that the motor operates at 500 r/min with a 1/3 rated load (0.33 N·m), and the current reference steps from 0.93 A (the corresponding current under the load) to 4.8 A (the maximum current).

Fig. 14(a) and (b) illustrate the situation with a mismatched inductance of 5.5 mH when the voltage constraint compensation is utilized and not utilized, respectively. Due to the large current

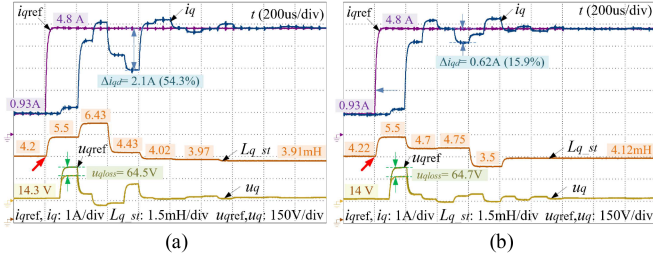


Fig. 14. q -axis current step response using the STIM-PCC method. (a) Without voltage constraint compensation. (b) With voltage constraint compensation.

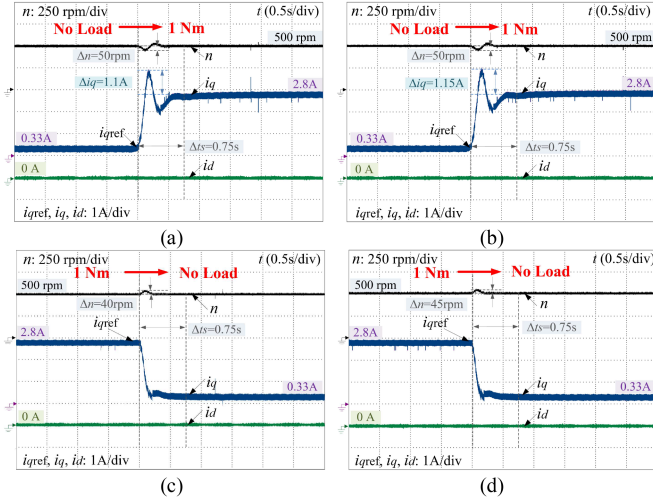


Fig. 15. Experimental results under torque variation conditions. (a) Load increases from no load to 1 N·m with conventional IM-PCC strategy. (b) Load increases from no load to 1 N·m with proposed STIM-PCC strategy. (c) Load decreases from 1 N·m to no load with conventional IM-PCC strategy. (d) Load decreases from 1 N·m to no load with proposed STIM-PCC strategy.

step, the u_{qref} for the second period after the step moment exceeds the limitation value and is constrained, resulting in a u_{qloss} of 64.5 V. When the compensation is not implemented, a significant drop in current (2.1 A, 54.3% of the step reference) during the dynamic process can be observed. At the same time, it can be seen that the corresponding nominal inductance is mistakenly tuned to be larger rather than smaller than the initial mismatched value, consistent with the deduction provided in Section III. In this case, the mismatched inductance is tuned within $11T_s$, with a relative error of 6.9%. However, by employing the proposed voltage compensation method, the excessive nominal inductance can be adjusted to a lower value at the first instance, and only $8T_s$ are required to tune the inductance with a relative error of 1.9%, significantly improving the tuning accuracy. Furthermore, the current drop is greatly reduced, and smooth tracking with fewer fluctuations is achieved, which proves the effectiveness of the proposed voltage constraint compensation method for O-M region.

3) *Speed Dynamic Performance With Sudden Load Changing*: In order to present the speed transient performance, Fig. 15 shows the experimental results with IM-PCC method and STIM-PCC method under the torque variation conditions, where a sudden torque variation of 1 N·m is introduced in the steady-state with the speed of 500 r/min.

In Fig. 15, since the same setting for the speed controller is employed in both control schemes, the difference between the speed responses with both methods is not significant. In Fig. 15(a) and (b), with a sudden load increase, the speed declines and i_{qref} rises to attain a heightened electromagnetic torque correspondingly. Through the regulation of speed controller, i_{qref} stabilizes amid fluctuations. Similarly, as the load decreases from 1 N·m to no load, as shown in Fig. 15(c) and (d), the speeds in both cases promptly increase to about 540 r/min and recovers to 500 r/min. Throughout these processes, the current reference is well tracked, signifying the great current control performance of the incremental model. Therefore, in the proposed control system, the small speed drop and rise resulting from the torque variation are caused by the torque variation, and the steady-state can be reconstructed quickly. Thus, with the proposed control scheme, the great speed dynamic performance can be achieved.

B. Steady-State Performance Verification

In this part, the steady-state performance of the proposed control scheme is evaluated by testing the performances at different operating speeds with initial inductance mismatch, verifying the broad effectiveness of the STIM-PCC strategy.

The steady-state test is conducted under specific conditions: the motor initially operates at 500 r/min with a rated load of 1 N·m. After 1 s, a specified value is set for the nominal q -axis inductance in the predictive model. Subsequently, after another second, the speed reference transitions from 500 to 1000 r/min (the rated speed). In the experiments, three values of the specified q -axis inductance—3, 5, and 4.2 mH—are chosen to represent the scenarios where the utilized nominal inductance is smaller, larger, or matched, respectively.

The results of these cases are depicted in Fig. 16.

As depicted in Fig. 16, when the nominal inductance is mismatched, through the designed self-tuning algorithm, the inductance mismatch can be swiftly diagnosed and tuned to a precise value. Throughout the entire process, irrespective of various operating conditions, such as reference steps in speed and current, the accurate tuning is consistently maintained, showcasing the strong robustness and wide adaptivity of the proposed STIM-PCC strategy. Additionally, it is evident that smooth and accurate control of the speed and d/q -axis currents is achieved with minimal fluctuations and excellent tracking performance across all operating conditions.

Furthermore, to evaluate the stator current quality in both steady states, a fast Fourier transform (FFT) algorithm is employed to analyze the total harmonic distortions (THDs) of the A-phase current (after inductance tuning). For the FFT analysis, the fundamental frequencies at 500 and 1000 r/min are 41.667 and 83.333 Hz, respectively, with a maximum frequency of 10 kHz. At each speed, six consecutive electrical cycles are selected and analyzed. According to the THD under each case, the steady-state current harmonics of distinct initial q -axis inductance mismatches with the proposed method can be compared, which are summarized in Table IV.

The table demonstrates that the THDs for all cases are consistently below 5%, signifying a low level of current harmonics and

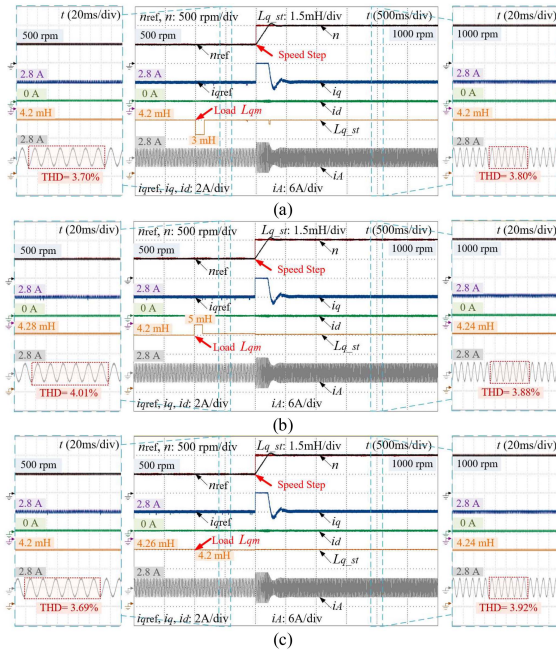


Fig. 16. Steady-state examination with the proposed STIM-PCC strategy. (a) $L_{qm} = 3 \text{ mH}$ ($\approx 0.7L_q$). (b) $L_{qm} = 5 \text{ mH}$ ($\approx 1.2L_q$). (c) $L_{qm} = 4.2 \text{ mH}$ (matched).

TABLE IV
STEADY-STATE CURRENT HARMONICS COMPARISON

L_{qm}	3mH ($0.7L_q$)		5mH ($1.2L_q$)		4.2mH (L_q)	
	L_{gst}	THD	L_{gst}	THD	L_{gst}	THD
500 rpm	4.2mH	3.70%	4.28mH	4.01%	4.2mH	3.69%
1000 rpm	4.2mH	3.80%	4.24mH	3.88%	4.24mH	3.92%

showcasing the excellent steady-state performance of the STIM-PCC scheme. Specifically, minimal differences are observed in the THDs when the tuned nominal inductances converge to different values. For instance, with an initial mismatched nominal inductance of 5 mH, at 500 r/min, the nominal inductance is tuned to 4.28 mH (approximately 2% higher than the actual value). In this case, the THD is 4.01%. When the nominal inductance is tuned to 4.2 mH (a matched value), the THD at the same speed is approximately 3.8%, slightly lower than the former case. The subtle difference in THD is attributed to the minimal current differentials at the steady-state, as indicated by the predictive model in (27), which results in a slight impact of inductance mismatch on the current harmonics. As a result, the THD variances between scenarios with different tuned values are not significant.

Summarily, to provide a comprehensive comparison, current transient performances and the steady state current harmonics with the proposed STIM-PCC method and conventional IM-PCC method are organized and compared as presented in Table V. In the table, the current overshoot (Δi_q) and 5% settling time (t_s) during the 2A-step responses with initially mismatched nominal inductance (as shown in part A), and the steady-state current harmonics at the rated condition (as shown in part B) are utilized as key indicators for the comparison.

TABLE V
CURRENT PERFORMANCE COMPARISON

L_{qm}	Method	Transient		Steady State
		Δi_q	t_s	THD
3.0mH	IM-PCC	0.32A	$10T_s$	3.82%
	STIM-PCC	0A	$7T_s$	3.80%
5.0mH	IM-PCC	0.37A	$75T_s$	4.23%
	STIM-PCC	0.38A	$8T_s$	3.88%
4.2mH	IM-PCC	0.11A	$6T_s$	3.75%
	STIM-PCC	0A	$6T_s$	3.92%

Table V underscores the significant enhancement in transient performance achieved through the implementation of the proposed STIM-PCC strategy. The comparison with the conventional IM-PCC method reveals that the proposed strategy results in reduced current fluctuations and shorter settling times during transient processes. Furthermore, the steady-state current harmonics across all cases remain below 5%, demonstrating the impeccable steady-state performance attained by the incremental predictive model. Besides, as aforementioned, since the current harmonics is slightly affected by the inductance mismatch, there exist minimal THD variances between cases with distinct nominal inductance.

Therefore, based on the aforementioned experimental results, it can be confirmed that the proposed STIM-PCC strategy offers substantial improvement in parameter robustness and transient performance compared to the conventional IM-PCC method. Additionally, the excellent control performance in the steady-state is also ensured in the proposed control scheme.

V. CONCLUSION

This article proposes a novel STIM-PCC strategy to improve the parameter robustness and current transient performance of the control system. In the conventional incremental model based predictive current control scheme, the reliance on the inductance makes the control system vulnerable to inductance mismatch, which compromises robustness and dynamic performance. To tackle this challenge, the STIM-PCC method is proposed, comprising three key modules. First, a mismatch criterion based on the prediction error is developed, by which the deviations between the nominal and actual inductance can be effectively identified. Second, a self-tuning algorithm is proposed, where the mismatched inductance is tuned to the actual value. Subsequently, according to the tuned inductance, the incremental predictive model is updated adaptively with the designed updating sequence. Furthermore, the tuning in the overmodulation region is considered, and a voltage constraint compensation method is put forward to enhance the tuning accuracy. The main contributions of the proposed work are summarized as follows.

- 1) Parameter robustness is greatly improved through the STIM-PCC method, where the inductance mismatch can be diagnosed and accurately tuned, with the independence of permanent magnet flux and resistance maintained.
- 2) The current transient performance under inductance mismatch is effectively enhanced, where not only the current

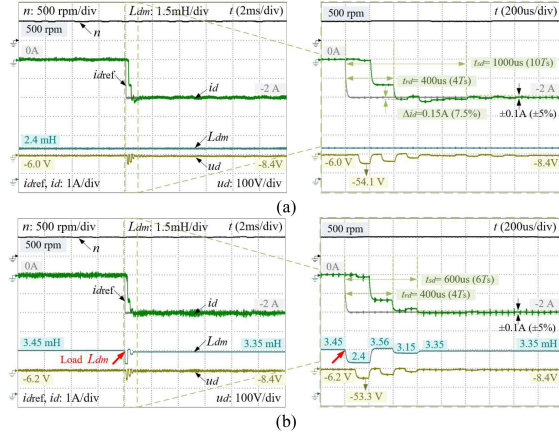


Fig. 17. d -axis current step response with inductance mismatch when $L_{dm} = 2.4$ mH ($\approx 0.7L_d$). (a) Conventional IM-PCC method. (b) Proposed STIM-PCC method.

settling time is significantly shortened, but the current fluctuations and oscillations are suppressed.

- 3) The overmodulation region is also considered in the STIM-PCC method, and through the voltage constraint compensation, the current drop during the dynamic process is observably reduced, increasing the adaptivity.
- 4) No extra control parameter is introduced in the proposed method, and complex calculation is avoided, where the control algorithm is simple to implement.

APPENDIX

A. Tuning Performance With Mismatched L_d

Similar to the examinations with mismatched L_q , the current step response under inductance mismatch is still utilized to illustrate the current transient performance and parameter robustness. In this part, the condition of the current step response test is that the motor runs at 500 r/min with rated load, and the d -axis current reference steps from 0 to -2 A.

When adopting the conventional IM-PCC method, the nominal inductance is set as the specified value. When employing the proposed STIM-PCC method, in order to depict the tuning process of the nominal inductance, the nominal inductance will be set as the same specified value at the “step instant,” and then be adjusted through the tuning algorithm. Similarly, two cases are taken as examples, where the L_{dm} are 2.4 mH ($\approx 0.7L_d$) and 4.14 mH ($\approx 1.2L_d$), respectively. The experimental results under both cases with the STIM-PCC and conventional IM-PCC methods are shown as follows.

As shown in Fig. 17, when the initial nominal inductance is smaller than the actual value, the proposed STIM-PCC strategy demonstrates superior dynamic performance compared to the conventional IM-PCC strategy. With the conventional IM-PCC strategy, noticeable overshoot is observed ($\Delta i_d = 0.15$ A, 7.5% of the current step), and it takes ten switching periods to settle the current ($t_s = 10T_s$). By adopting the proposed STIM-PCC strategy, the inductance mismatch is quickly detected and adjusted

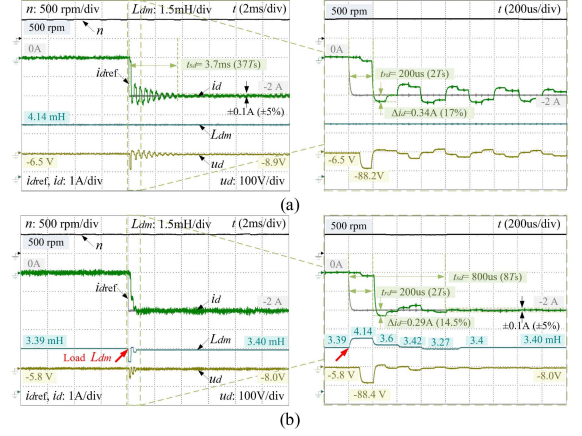


Fig. 18. d -axis current step response with inductance mismatch when $L_{dm} = 4.14$ mH ($\approx 1.2L_d$). (a) Conventional IM-PCC method. (b) Proposed STIM-PCC method.

at the second control instant after setting the mismatched inductance, thus the current overshoot can be effectively eliminated. Compared to the conventional IM-PCC method, the settling time is also reduced significantly, with only 6T_s required to settle the current. Besides, as the tuning process shows, the mismatched inductance can be tuned in just 6T_s, with a relative error of only 2.90%, demonstrating the high accuracy of the proposed self-tuning algorithm.

Similarly, as shown in Fig. 18, when the initial nominal inductance is larger than the actual value. While using the conventional IM-PCC method, noticeable overshoot is observed ($\Delta i_d = 0.34$ A, 17% of the current step), and the d -axis current follows the reference with long-term fluctuations, and it takes 37 switching periods to settle the current ($t_s = 37T_s$). However, with the proposed STIM-PCC strategy, the current settling time is significantly reduced, requiring only 8T_s (a reduction of 29T_s). Besides, precise inductance tuning is achieved, with the mismatched inductance being tuned in just 9T_s and a relative error of only 1.45%.

The above results strongly demonstrate the effectiveness of the proposed STIM-PCC method, where reduced current fluctuations during the dynamic process and significantly shortened settling time can be achieved, and inductance mismatch can be swiftly detected and accurately tuned.

B. Evaluations on the Impact of Current Sampling Frequency

Typically, currents are sampled at the beginning of the switching period, with the current sampling frequency matching the switching frequency. To evaluate the self-tuning performance across various current sampling frequencies, experiments were conducted. The current step response under inductance mismatch is still adopted to showcase the current transient performance and self-tuning performance.

The experiments covered a switching frequency range from 5 to 15 kHz, with five frequencies chosen at 2.5 kHz intervals (15, 12.5, 10, 7.5, and 5 kHz), and the results are presented in Fig. 19. Each test is conducted with the motor operating at 500 rpm under the rated load, with the current reference stepping from 2.8 to 4.8 A. In order to depict the tuning process

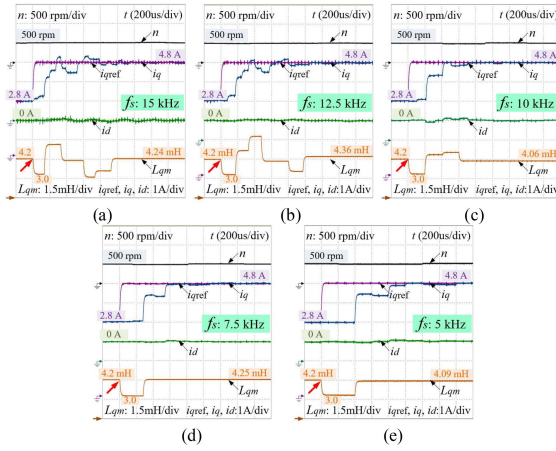


Fig. 19. Current step responses with different current sampling frequencies when $L_{qrm} = 3 \text{ mH}$ ($\approx 0.7L_q$). (a) $f_s = 15 \text{ kHz}$. (b) $f_s = 12.5 \text{ kHz}$. (c) $f_s = 10 \text{ kHz}$. (d) $f_s = 7.5 \text{ kHz}$. (e) $f_s = 5 \text{ kHz}$.

TABLE VI
SELF-TUNING ERRORS UNDER DIFFERENT FREQUENCIES

f_s	15kHz	12.5kHz	10kHz	7.5kHz	5kHz
Self-Tuning Error	0.95%	3.81%	3.33%	1.19%	2.62%

of the nominal inductance, the nominal q -axis inductance in the predictive model is deliberately set to a mismatched value ($0.7L_q$) at the “step instant.” Subsequently, it is adjusted through the tuning algorithm.

Fig. 19 illustrates that in all scenarios, the initially mismatched inductance is promptly detected and tuned within 1 ms. Particularly, with higher frequencies such as 15 and 12.5 kHz, the q -axis current swiftly rises to track the reference with minor fluctuations. Conversely, situations with lower frequencies like 7.5 and 5 kHz exhibit a longer current rise time, yet the reference tracking is smooth without overshoot. Besides, the results reveal that under each current sampling frequency, self-tuning errors remain below 4%, highlighting the effectiveness of the proposed self-tuning algorithm.

According to Fig. 19, self-tuning performances of all cases with the proposed control method can be compared, the tuning errors under each frequency are summarized in Table VI.

Based on these experimental findings, it is evident that the proposed control strategy owns strong adaptability across a broad current sampling frequency range, underscoring the versatility of the proposed control approach.

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